

GRAD-E1487: Causal Machine Learning**1. General information**

Course Format	Onsite
Instructor(s)	Drew Dimmery
Instructor's E-mail	d.dimmery@hertie-school.org
Assistant (if applicable)	Ubaida Ansari (u.ansari@hertie-school.org)
Instructor's Office Hours	Mondays, 1-3pm, or by appointment.

Link to [Study, Examination and Admission Rules and MIA, MDS and MPP Module Handbooks](#)

For information on **course room**, times, and **session dates**, please consult the [Course Plan](#) on *MyStudies*.

Instructor Information:

Drew Dimmery is Professor of Data Science for the Common Good at the Hertie School. His research interests include causal inference, machine learning, and the usage of data science for public policy. Prior to moving back to academia, he was a research scientist at Facebook in the United States for four years. He holds a PhD in political methodology from New York University and has published in top machine learning journals such as ICML, KDD and AISTATS and general interest journals like Science, Nature and Proceedings of the National Academy of Sciences.

2. Course Contents and Learning ObjectivesCourse contents:

This course is an introduction and survey of causal machine learning. It examines a wide variety of these approaches and how they can be used in the course of doing causal research in the social sciences. It presumes an existing knowledge of both causal inference and of machine learning. We will read a selection of papers introducing and implementing techniques and research designs, with applications to the social and health sciences and public policy. We will discuss what these new methods are able to offer, and where they may have limitations. The course will be oriented around class discussion and student presentations on the readings. Students will further gain the ability to critically evaluate methodologies through the use of Monte Carlo simulations.

Learning objectives:

Students will:

- Learn to use simulations to understand the mechanics of a method/estimator
- Understand how to use machine learning methods for causal questions in the social sciences
- Become proficient in applying these methods to problems

The course is oriented around class discussion and student presentations, so students are expected to take an active role.

Prerequisites:

- Has taken Machine Learning
- Has taken Causal Inference
- Comfortable training ML models in Python using sklearn, PyTorch, doing numerical computing using NumPy

Diversity Statement:

Understanding and respect for all cultures and ethnicities is central to the teaching at Hertie. Being mindful of diversity is an important issue for policy professionals in the planning, implementation, and evaluation of programs designed for specific groups, populations, or communities. Diversity and cultural awareness will be integrated in the course content whenever possible.

3. Grading and Assignments

Composition of Final Grade:

Assignment 1: Referee report	Before corresponding class	Submit via GitHub	15%
Assignment 2: Class presentation	Organized on first day of class	In-class	20%
Assignment 3: Class demonstration	Organized on first day of class	In-class	25%
Assignment 4: Project	End of term (December 19)	Submit via GitHub	30%
Participation grade	-	-	10%

Assignment Details

Assignment 1: Referee Report

Select one of the papers on the syllabus, including from the additional references (but not the "foundation" paper). Write a referee report as if you were requested to review the paper for publication at a journal, summarizing the paper's main contributions, areas that could be strengthened, problems, and additional extensions or analyses that you would like to see. I expect a concise but information-dense report of 1-2 pages. Submit the referee report on GitHub 36 hours before the paper is discussed in class. The paper selected for the referee report cannot be from the same class as the class presentation or demonstration.

References on how to write a referee report:

- Berk, J., C. R. Harvey, and D. A. Hirshleifer (2016). Preparing a referee report: guidelines and perspectives. Available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2547191
- <https://chrisblattman.com/blog/2012/01/18/how-to-referee-an-academic-paper/>

Preparation of this assignment should take no more than a few hours.

Assignment 2: Class Presentation

Each student will be responsible for presenting and leading discussion on one of the required papers, beginning the third week of the term. Discussion leaders will provide a 10-15 minute high-level overview of the paper presented as a slide-deck, and will develop a list of discussion questions for the class to discuss. For classes with multiple leaders, co-discussion leaders should divide papers among themselves but may not discuss the same paper. Discussion questions should be submitted to the instructor by email the evening before class. Students should come to the Professor's office hours at least one week before they are scheduled to present to discuss the paper and ensure they understand everything they need to understand about their assigned paper.

Preparation of this presentation should take 4-6 hours. In contrast to the referee report, students should gain a sufficiently thorough understanding of the paper that they can explain it to their classmates effectively.

Assignment 3: Class Demonstration

Each student will be responsible for providing a practical demonstration of one week's topic beginning the third week of the term. This will be in the form of a fully functioning Rmarkdown or Jupyter notebook (it should run from start to finish with no errors). The student will lead the class in walking through the notebook, demonstrating the code and the output of the notebook. This demonstration can be one of a variety of things:

1. Running a method discussed during the week on an applied problem of interest (e.g. a dataset of interest)
2. Implementing the method "from scratch" and highlighting how it works and specific implementation-specific concerns that must be addressed
3. A simulation detailing the performance of the method (or multiple methods) in a setting where the "truth" is known

Whichever approach the student takes, this should take approximately 20 minutes. The notebook should (perhaps by reducing the number of replicates) be able to be run within a few minutes in-class.

If the class-size demands that multiple students are assigned to a single class, then they will confer with the professor about how to split up the week's session or work together.

Preparation of this demonstration should take 1-2 days of work. While the presentation focuses on understanding the paper, the demonstration ensures that the student has practical understanding of the implementation, how it can be used and/or what its strengths and weaknesses are when applied to data. Students should take a few iterations to ensure that the analysis provides insight into the methods under consideration in the session.

Assignment 4: Final Project

The final project is a work of exposition/tutorial in which you explore a topic related to machine learning for causal inference both analytically and computationally. Discuss the foundations for this work in statistics or related fields. Then, choose and focus on some aspect of how this work has developed, competing methodologies or debates with associated benefits and drawbacks, and directions for future research. Demonstrate these ideas with a computational example (i.e. write code). The deliverable is a 5 page paper that should include focused code snippets (if appropriate) and key output as well as a script or notebook and related data that can be run all the way through after setting the working directory. For example, you can write an Rmd or Jupyter notebook where only the most important code and output for understanding your point is shown in the pdf output (there should not be extraneous code in the report). The reader of your report should come away with a good understanding of the strengths and drawbacks of the method/theme as well as how to use them effectively. The class

demonstration is a good place to start for this project, but you are not *required* to have your project be an extension of your class demonstration.

You will be graded not only on the accuracy of your analysis, but also on the clarity of your exposition, which includes notation, typesetting (equation alignment, exceeding margins), grammar, figures, tables, etc. All code should be reproducible—please provide any supporting files necessary with your submission. The more difficult it is to read your work, the fewer points you will receive. Follow a coding style guide (e.g., <https://style.tidyverse.org/> and <https://peps.python.org/pep-0008/>) and stick to it. The expectation is for polished work.

Alternatively, students may submit original work, such as a research proposal for a project they are working on. This option is available with instructor permission only.

Preparation of this report should take up to a week of additional work between extending the analyses of the demonstration and bringing it up to the quality of presentation necessary for the report.

Participation grade

The participation grade assumes that students take part, not as passive consumers of knowledge, but as active participants in the exchange, production, and critique of ideas—their own ideas and the ideas of others. Therefore, students should come to class not only having read and viewed the materials assigned for that day but also prepared to discuss the readings and to contribute thoughtfully to the conversation. This includes reading their colleagues' response papers before class. Participation is marked by its active nature, its consistency, and its quality. To receive the best possible participation grades, students should ask thoughtful questions, build on their colleagues' comments, and come prepared to class with specific observations about the readings.

This class is structured as a seminar, so there will be substantial expectations on students to contribute. Students should spend a few hours ahead of class reading the assigned readings carefully. Active reading (i.e., highlighting and taking notes) will ensure that students are prepared to contribute in class.

Late submission of assignments: For each day the assignment is turned in late, the grade will be reduced by 10% (e.g., submission two days after the deadline would result in a 20% grade deduction).

Attendance: Students are expected to be present and prepared for every class session. Active participation during lectures and seminar discussions is essential. If unavoidable circumstances arise which prevent attendance or preparation, the instructor should be advised by email with as much advance notice as possible. Please note that students cannot miss more than two out of 12 course sessions. For further information, please consult the [Examination Rules](#) §10.

Academic Integrity: The Hertie School is committed to the standards of good academic and ethical conduct. Any violation of these standards shall be subject to disciplinary action. Plagiarism, misuse of AI, free riding in group work, and other deceitful actions are not tolerated. See [Examination Rules](#) §16, the Hertie [Plagiarism Policy](#), and [the Hertie Guidelines for Artificial Intelligence Tools](#).

Compensation for Disadvantages: If a student furnishes evidence that he or she is not able to take an examination as required in whole or in part due to disability or permanent illness, the Examination Committee may, upon written request, approve learning accommodation(s). In this respect, the submission of adequate certificates may be required. See [Examination Rules](#) §14.

Extenuating circumstances: An extension can be granted due to extenuating circumstances (i.e., for reasons like illness, personal loss or hardship, or caring duties). In such cases, please contact the course instructor and examination office *in advance* of the assignment deadline.

4. General Readings

There is no overall textbook for this course; required readings are all papers available online (and which will be posted on GitHub). A few texts may be useful to the student, but they are completely optional. If you're considering getting any of them, come by the professor's office to browse them first before committing to the sometimes high price.

Optional Textbooks:

- **Joshua D Angrist and Jörn-Steffen Pischke.** *Mostly harmless econometrics: An empiricist's companion*. Princeton university press, 2009 — *Compellingly written. It's a little dated, now, but still a great resource.*
- **Alan S Gerber and Donald P Green.** *Field experiments: Design, analysis, and interpretation*. WW Norton, 2012 — *Neither the most rigorous nor the most up-to-date book, but is probably still the best general-purpose handbook for experimentation.*
- **Miguel A Hernán and James M Robins.** *Causal inference*, 2010 — *This comes from an epidemiological perspective, but is a great overview of important causal and statistical concepts.*
- **Guido W Imbens and Donald B Rubin.** *Causal inference in statistics, social, and biomedical sciences*. Cambridge University Press, 2015 — *This is a great general-purpose reference book to have on hand for causal inference.*
- **Martin Huber.** *Causal Analysis: Impact Evaluation and Causal Machine Learning with Applications in R*, 2023 — *This is one of a very few books focusing on CausalML.*
- **Chernozhukov, Victor, Christian Hansen, Nathan Kallus, Martin Spindler and Vasilis Syrgkanis.** *Applied Causal Inference Powered by ML and AI*, 2025 — *This is the other CausalML book. The authors are great, but it doesn't quite make it to the level of a "canonical" text.*

5. Course Sessions and Readings

All course readings can be accessed on the course page. I will provide additional optional readings as appropriate.

Typical class organization:

Activity	Duration	Led by
Session introduction	10-20 minutes	Professor
Discussion of readings	30-45 minutes	One student per paper
Demonstrations	30-45 minutes	Assigned students

Session 1: Basics of Design-based Causal Inference / Monte Carlo Simulation	
9. September	
Required Readings	None
Optional Readings	• Imbens, G. W., & Rubin, D. B. (2015). <i>Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction</i>, Chapters 1-2. Cambridge University Press. • Murphy, J. R. (1997). How to read the statistical methods literature: a guide for students. <i>The American Statistician</i> 51(2), 155-157 • Morris, T. P., White, I. R., & Crowther, M. J. (2019). Using simulation studies to evaluate statistical methods. <i>Statistics in Medicine</i>, 38(11), 2074-2102.

Session 2: Covariate Adjustment: when you model the outcome

16. September

Required Readings	• Anastasios A Tsiatis, Marie Davidian, Min Zhang, and Xiaomin Lu. Covariate adjustment for two-sample treatment comparisons in randomized clinical trials: a principled yet flexible approach. <i>Statistics in medicine</i>, 27(23):4658-4677, 2008. • Aronow, P. M., & Middleton, J. A. (2013). A class of unbiased estimators of the average treatment effect in randomized experiments. <i>Journal of Causal Inference</i>, 1(1), 135-154. https://doi.org/10.1515/jci-2012-0009 • Lin, W. (2013). Agnostic notes on regression adjustments to experimental data: Reexamining Freedman's critique. <i>Annals of Applied Statistics</i>, 7(1), 295-318.
Optional Readings	• Belloni, A., Chernozhukov, V., & Hansen, C. (2014). "Inference on Treatment Effects after Selection among High-Dimensional Controls." <i>Review of Economic Studies</i>, 81(2), 608-650. • Bloniarz, A., H. Liu, C.-H. Zhang, J. S. Sekhon, and B. Yu (2016). Lasso adjustments of treatment effect estimates in randomized experiments. <i>Proceedings of the National Academy of Sciences</i> 113(27), 7383-7390 • Jennifer L Hill. Bayesian nonparametric modeling for causal inference. <i>Journal of Computational and Graphical Statistics</i>, 20(1): 217-240, 2011. • Foundation: Freedman, D. A. (2008). On regression adjustments to experimental data. <i>Advances in Applied Mathematics</i>, 40(2), 180-193.

Session 3: Balancing Weights: when you model the treatment

23. September

Required Readings	• Imai, K., & Ratkovic, M. (2014). "Covariate Balancing Propensity Score." <i>Journal of the Royal Statistical Society: Series B</i>, 76(1), 243-263. • David Arbour, Drew Dimmery, and Arjun Sondhi. Permutation weighting. In <i>International Conference on Machine Learning</i>, pages 331-341. PMLR, 2021C.
Optional Readings	• Fan Li, Laine E Thomas, and Fan Li. Addressing extreme propensity scores via the overlap weights. <i>American journal of epidemiology</i>, 188(1):250-257, 2019. • Zhao, Q. (2019). "Covariate Balancing Propensity Score by Tailored Loss Functions." <i>Annals of Statistics</i>, 47(2), 965-993. • Zubizarreta, J. R. (2015). "Stable Weights that Balance Covariates for Estimation With Incomplete Outcome Data." <i>Journal of the American Statistical Association</i>, 110(511), 910-922. • Hainmueller, J. (2012). Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. <i>Political Analysis</i>, 20(1), 25-46. • Foundation: Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. <i>Biometrika</i>, 70(1), 41-56.

Session 4: Double Robustness, Double ML and TMLE: when you model both outcome and treatment

30. September

Required Readings	• Kennedy, E. H. (2022). Semiparametric doubly robust targeted double machine learning: a review. arXiv preprint arXiv:2203.06469. Parts 1-3. • Chernozhukov, V., D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins (2018). Double/debiased machine learning for treatment and structural parameters: Double/debiased machine learning. The Econometrics Journal 21(1), C21 Parts 1-3. • Schuler, M. S. and S. Rose (2017). Targeted maximum likelihood estimation for causal inference in observational studies. American Journal of Epidemiology 185(1), 65-73
Optional Readings	• van der Laan, M. J., & Rose, S. (2011). "Targeted Learning: Causal Inference for Observational and Experimental Data", Chapters 4-6. Springer. • Kang, J. D., & Schafer, J. L. (2007). "Demystifying Double Robustness: A Comparison of Alternative Strategies for Estimating a Population Mean from Incomplete Data." Statistical Science, 22(4), 523-539. • Foundation: Bang, H., & Robins, J. M. (2005). "Doubly Robust Estimation in Missing Data and Causal Inference Models." Biometrics, 61(4), 962-973.

Session 5: Heterogeneous Treatment Effects I: Basics

7. October

Required Readings	• Wager, S. and S. Athey (2018). Estimation and inference of heterogeneous treatment effects using random forests. Journal of the American Statistical Association 113(523), 1228-1242 • Künzel, S. R., J. S. Sekhon, P. J. Bickel, and B. Yu (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. Proceedings of the national academy of sciences 116(10), 4156-4165 • Kennedy, E. H. (2020). Optimal doubly robust estimation of heterogeneous causal effects. arXiv preprint arXiv:2004.14497.
Optional Readings	• Athey, S. and G. Imbens (2016). Recursive partitioning for heterogeneous causal effects. Proceedings of the National Academy of Sciences 113(27), 7353-7360 • Athey, S., J. Tibshirani, and S. Wager (2019). Generalized random forests • Nie, X., & Wager, S. (2021). Quasi-oracle estimation of heterogeneous treatment effects. Biometrika, 108(2), 299-319. • Foundation: Holland, P. W. (1986). Statistics and causal inference. Journal of the American Statistical Association, 81(396), 945-960.

Session 6: Heterogeneous Treatment Effects II: Neural networks

14. October

Required Readings	• Nie, X., & Wager, S. (2021). "Quasi-oracle Estimation of Heterogeneous Treatment Effects." Biometrika, 108(2), 299-319. • Shi, C., Blei, D., & Veitch, V. (2019). "Adapting Neural Networks for the Estimation of Treatment Effects." Advances in Neural Information Processing Systems, 32.
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Optional Readings	• Alaa, A. M., & van der Schaar, M. (2017). "Bayesian Inference of Individualized Treatment Effects using Multi-task Gaussian Processes." Advances in Neural Information Processing Systems, 30. • Yoon, J., Jordon, J., & van der Schaar, M. (2018). GANITE: Estimation of individualized treatment effects using generative adversarial nets. International Conference on Machine Learning, 5101-5110. • Louizos, C., Shalit, U., Mooij, J. M., Sontag, D., Zemel, R., & Welling, M. (2017). Causal effect inference with deep latent-variable models. Advances in Neural Information Processing Systems, 30. • Ma, Y., Frauen, D., Javurek, E., & Feuerriegel, S. (2025). Foundation Models for Causal Inference via Prior-Data Fitted Networks. arXiv:2506.10914 • Foundation: Shalit, U., Johansson, F. D., & Sontag, D. (2017). "Estimating Individual Treatment Effect: Generalization Bounds and Algorithms." International Conference on Machine Learning, 3076-3085.
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Session 7: Off-policy evaluation and optimization: Do you need causal effects?

21. October

Required Readings	• Athey, S. and S. Wager (2021). Policy learning with observational data. Econometrica 89(1), 133-161 • Fernández-Loría, C., & Provost, F. (2022). "Causal Decision Making and Causal Effect Estimation Are Not the Same...and Why It Matters." INFORMS Journal on Data Science, 1(1), 4-16. • Luedtke, A. R., & van der Laan, M. J. (2016). Super-learning of an optimal dynamic treatment rule. International Journal of Biostatistics, 12(1), 305-332.
Optional Readings	• Kern, C., Fischer-Abaigar, U., Schweisthal, J. et al. Algorithms for reliable decision-making need causal reasoning. Nat Comput Sci 5, 356-360 (2025). https://doi.org/10.1038/s43588-025-00814-9 • Swaminathan, A., & Joachims, T. (2015). Counterfactual risk minimization: Learning from logged bandit feedback. International Conference on Machine Learning, 814-823. • Jiang, N., & Li, L. (2016). Doubly robust off-policy value evaluation. International Conference on Machine Learning, 652-661. • Kallus, N. (2018). "Balanced Policy Evaluation and Learning." Advances in Neural Information Processing Systems, 31. • Kitagawa, T., & Tetenov, A. (2018). Who should be treated? Empirical welfare maximization methods for treatment choice. Econometrica, 86(2), 591-616. • Foundation: Dudík, M., Langford, J., & Li, L. (2011). "Doubly Robust Policy Evaluation and Learning." International Conference on Machine Learning, 1097-1104.

Session 8: Experimental Design: How should you collect data to make your life easier?

4. November

Required Readings	• Nathan Kallus. Optimal a priori balance in the design of controlled experiments. Journal of the Royal Statistical Society: Series B (Statistical Methodology), 80(1):85-112, 2018.
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	• Harshaw, C., Sävje, F., Spielman, D., & Zhang, P. (2019). "Balancing Covariates in Randomized Experiments with the Gram-Schmidt Walk Design." arXiv preprint arXiv:1911.03071 • Morgan, K. L., & Rubin, D. B. (2012). Rerandomization to improve covariate balance in experiments. <i>Annals of Statistics</i>, 40(2), 1263-1282.
Optional Readings	• Arbour, D., Dimmery, D., Mai, T., & Rao, A. (2022). "Online Balanced Experimental Design." <i>Proceedings of the 39th International Conference on Machine Learning (ICML)</i>, 844-864 • Arbour, D., Dimmery, D., & Rao, A. (2021). Efficient balanced treatment assignments for experimentation. <i>Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (AISTATS)</i>, 3070-3078. • Michael J Higgins, Fredrik Sävje, and Jasjeet S Sekhon. Improving massive experiments with threshold blocking. <i>Proceedings of the National Academy of Sciences</i>, 113(27):7369-7376, 2016. • Foundation: Greevy, R., Lu, B., Silber, J. H., & Rosenbaum, P. (2004). Optimal multivariate matching before randomization. <i>Biostatistics</i>, 5(2), 263-275.

Session 9: Panel data and 🍁 the new Diff-in-diff 🍁 : Using time for identification

11. November

Required Readings	• Clément De Chaisemartin and Xavier d'Haultfoeuille. Two-way fixed effects estimators with heterogeneous treatment effects. <i>American Economic Review</i>, 110(9):2964-2996, 2020. • Andrew Goodman-Bacon. Difference-in-differences with variation in treatment timing. <i>Journal of Econometrics</i>, 225(2):254-277, 2021. • Brantly Callaway and Pedro HC Sant'Anna. Difference-in-differences with multiple time periods. <i>Journal of Econometrics</i>, 225(2): 200-230, 2021. • Ben-Michael, E., Feller, A., & Rothstein, J. (2021). "The Augmented Synthetic Control Method." <i>Journal of the American Statistical Association</i>, 116(536), 1789-1803
Optional Readings	• Sun, L., & Abraham, S. (2021). "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." <i>Journal of Econometrics</i>, 225(2), 175-199. • Ben-Michael, E., Arbour, D., Feller, A., Franks, A., & Raphael, S. (2023). "Estimating the Effects of a California Gun Control Program with Multitask Gaussian Processes." <i>Annals of Applied Statistics</i>, 17(2), 985-1016 • Foundation: Abadie, A., Diamond, A., & Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. <i>Journal of the American Statistical Association</i>, 105(490), 493-505.

Session 10: Partial Identification: when you don't have the ideal data for the task

18. November

Required Readings	• Yadlowsky, S., Namkoong, H., Basu, S., Duchi, J., & Tian, L. (2022). "Bounds on the Conditional and Average Treatment Effect with Unobserved Confounding Factors." <i>Annals of Statistics</i>, 50(5), 2587-2615
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	• Samii, C., Wang, Y., & Zhou, J. (2023). "Generalizing Trimming Bounds for Endogenously Missing Outcome Data Using Random Forests." arXiv preprint arXiv:2309.08985 • Khan, S., Saveski, M., & Ugander, J. (2024). "Off-Policy Evaluation Beyond Overlap: Partial Identification Through Smoothness." Proceedings of the 41st International Conference on Machine Learning (ICML)
Optional Readings	• Tamer, E. (2010). "Partial Identification in Econometrics." Annual Review of Economics, 2(1), 167-195. • Kallus, N., Mao, X., & Uehara, M. (2022). "Causal Inference Under Unmeasured Confounding With Negative Controls: A Minimax Learning Approach." Journal of the American Statistical Association, 117(540), 2071-2086. • Imbens, G. W., & Manski, C. F. (2004). Confidence intervals for partially identified parameters. Econometrica, 72(6), 1845-1857. • Foundation: Manski, C. F. (1990). Nonparametric bounds on treatment effects. American Economic Review, 80(2), 319-323.

Session 11: Response Adaptation and reinforcement learning: when you can interact rather than merely act

25. November

Required Readings	• Russo, D., Van Roy, B., Kazerouni, A., Osband, I., & Wen, Z. (2018). "A Tutorial on Thompson Sampling." Foundations and Trends in Machine Learning, 11(1), 1-96. [Read Sections 1-3] • Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction, Chapters 1-3, 13 (Policy Gradient Methods). MIT Press. • Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). "Proximal Policy Optimization Algorithms." arXiv preprint arXiv:1707.06347 • Haarnoja et al. (2018). "Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor." ICML.
Optional Readings	• Hadad, V., Hirshberg, D. A., Zhan, R., Wager, S., & Athey, S. (2021). "Confidence Intervals for Policy Evaluation in Adaptive Experiments." PNAS, 118(15). • Ouyang, L., Wu, J., Jiang, X., et al. (2022). "Training Language Models to Follow Instructions with Human Feedback." Advances in Neural Information Processing Systems, 35, 27730-27744 • Kasy, M., & Sautmann, A. (2021). Adaptive treatment assignment in experiments for policy choice. Econometrica, 89(1), 113-132. • Ariu, K., Kato, M., Komiyama, J., McAlinn, K., & Qin, C. (2021). Policy choice and best arm identification: Asymptotic analysis of exploration sampling. arXiv preprint arXiv:2109.08229. • Foundation: Robbins, H. (1952). Some aspects of the sequential design of experiments. Bulletin of the American Mathematical Society, 58(5), 527-535.

Session 12: Interference: What if units are not independent from one another?

2. December

Required Readings	• Aronow, P. M., & Samii, C. (2017). "Estimating Average Causal Effects under General Interference, with Application to a Social Network Experiment." <i>Annals of Applied Statistics</i>, 11(4), 1912-1947 • Ugander, J., Karrer, B., Backstrom, L., & Kleinberg, J. (2013). "Graph Cluster Randomization: Network Exposure to Multiple Universes." <i>Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining</i>, 329-337. • Wager, S., & Xu, K. (2021). "Experimenting in Equilibrium." <i>Management Science</i>, 67(11), 6694-6715.
Optional Readings	• Sävje, F., Aronow, P. M., & Hudgens, M. G. (2021). "Average Treatment Effects in the Presence of Unknown Interference." <i>Annals of Statistics</i>, 49(2), 673-701. • Shirani, S., & Bayati, M. (2024). "Causal Message-Passing for Experiments with Unknown and General Network Interference." <i>Proceedings of the National Academy of Sciences</i>, 121(40), e2322232121 • Foundation: Hudgens, M. G., & Halloran, M. E. (2008). Toward causal inference with interference. <i>Journal of the American Statistical Association</i>, 103(482), 832-842.

This syllabus is subject to change at the instructor's discretion. All updates will be communicated via Moodle and in class.